

Tree growth responses to extreme drought events are not well predicted by climate

Victor Van der Meersch¹, Benjamin I. Cook^{2,3}, Michael Betancourt⁴, E.M. Wolkovich¹

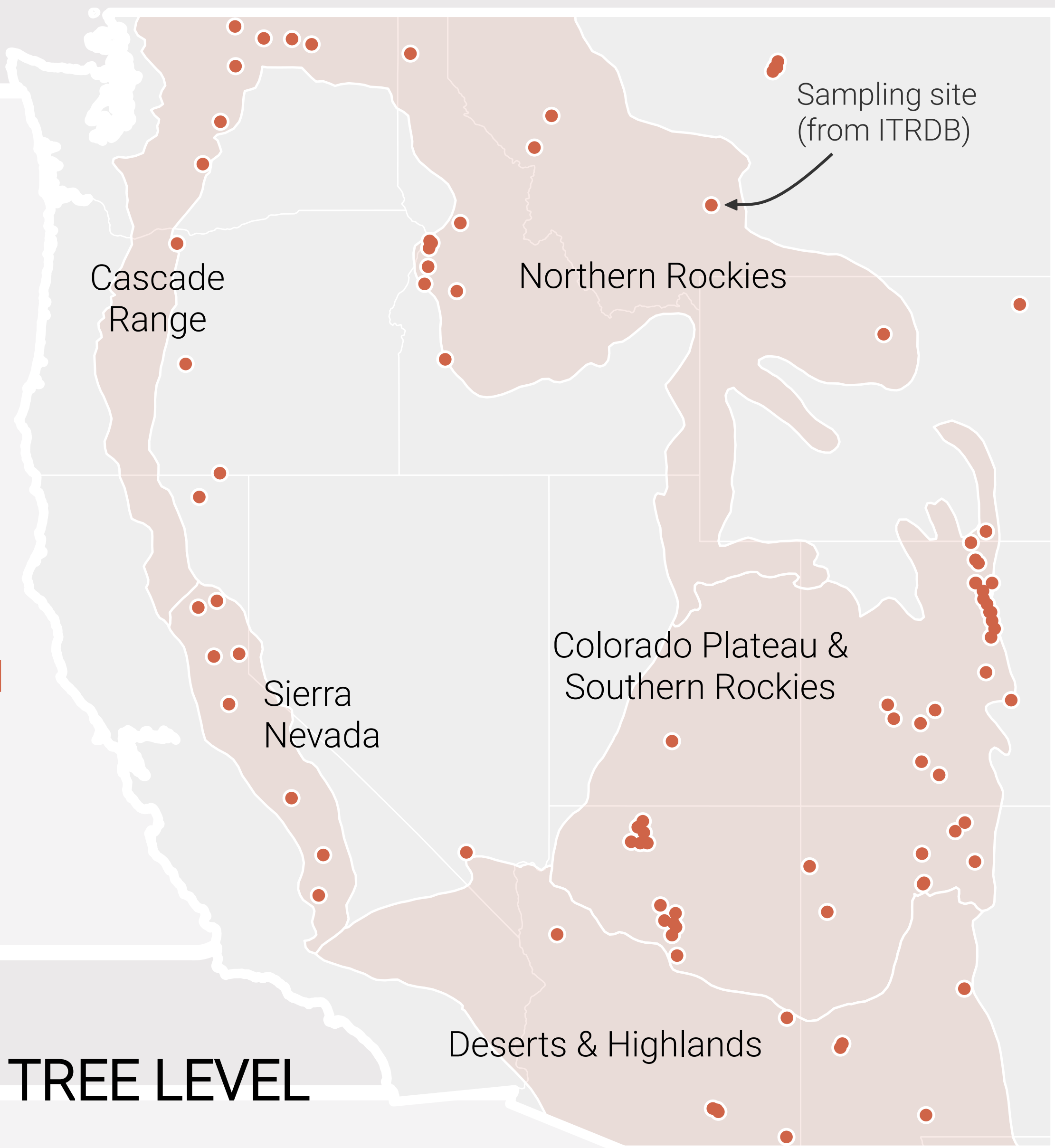
¹ University of British Columbia, Vancouver, Canada
² Goddard Institute for Space Studies, NASA, New York, USA
³ Lamont-Doherty Earth Observatory, Columbia University, Palisades, USA
⁴ Symplectomorphic LLC, New York, USA

OVERVIEW

- Climate change is **intensifying droughts** through shifts in temperature, vapor pressure deficit (VPD), and soil moisture [1]
- Combined effects of these drivers on tree growth remain poorly understood at large scales [2,3]
- A new **Bayesian approach** to jointly model tree growth across Western US, at the tree level
- Our model is able to separate climate effects from other complex dynamics (e.g. allometry)
- Some extreme drought caused **exceptional growth reductions** far exceeding what standard climate variables predict
- Current models likely **underestimate growth declines** during extreme events

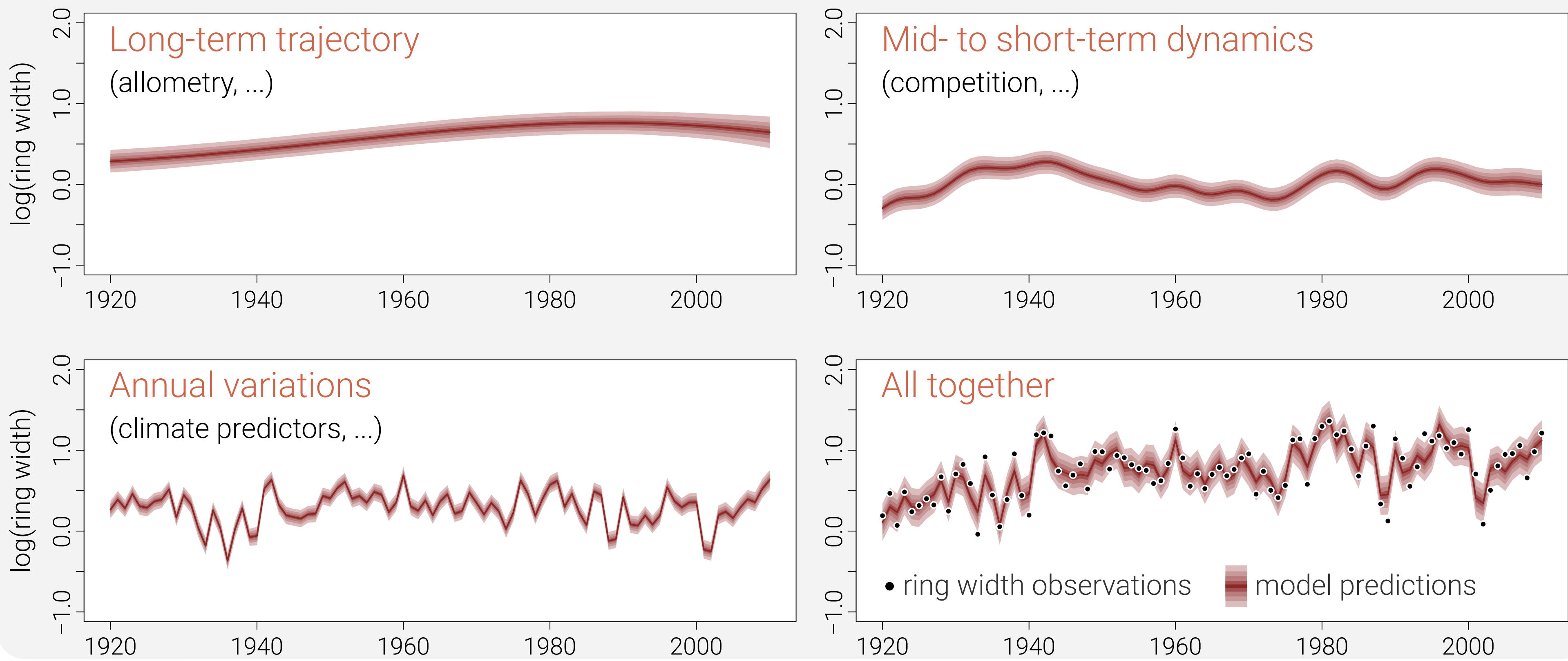
1. BACKGROUND

- Tree-ring width time series are **one of the longest and most spatially extensive** records of forest response to climate
- However, tree rings are affected by **many drivers other than climate**: age-size effects, competition, biotic disturbance...
- Standard dendrochronology approaches detrend individual trees and aggregate them to identify some shared climatic signal [4]
- But **drought stress operates at the individual level**, whose responses integrate non-linear physiological processes [5]
- To capture how extreme events, we need to model individual trees explicitly



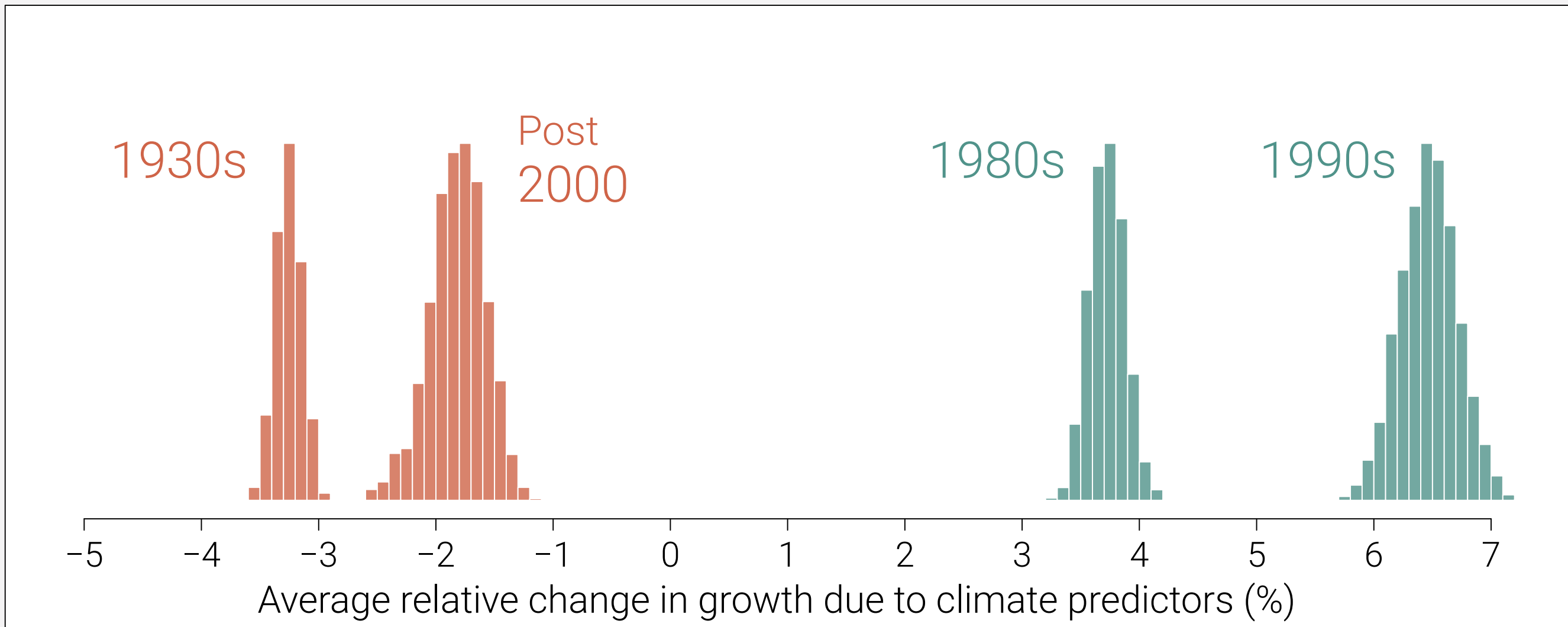
2. MODELING GROWTH AT THE TREE LEVEL

Tree growth results from three major components that we jointly model:



3. MAIN CLIMATIC DRIVERS OF GROWTH

Growing degree days, winter precipitation, and vapor pressure deficit lead on average to **±10% of changes in ring width**.

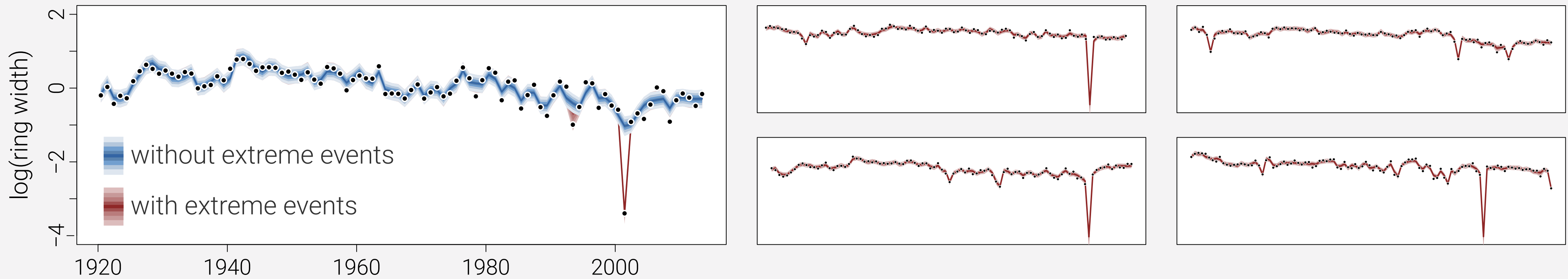


Our climate-growth component generally captures some known **wet periods** and some **less favorable periods** for growth.

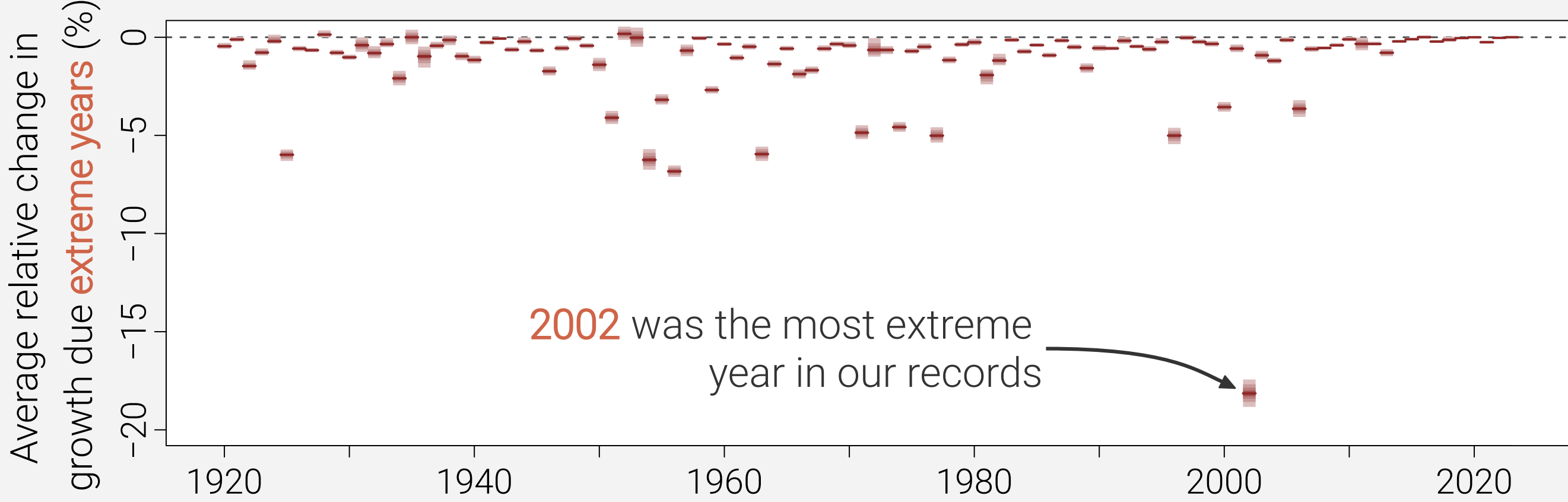
4. EXTREME GROWTH REDUCTIONS (SHOCKS)

But a tree can show **severe growth suppression**, far beyond what its climate response predicts.

When one tree experiences a shock, **neighboring trees within the same forest** tend to as well.



These shocks have major **consequences at the ecosystem scale**.



The synchrony of shocks reveals a shared, forest-level signal, consistent with a **common external stressor** (rather than individual chance).

Many of these shocks correspond to known major drought events, which **suppress growth by up to almost 20%** on top of the effects of climate predictors.

5. IMPLICATIONS FOR FORESTS AND FORECASTS

- Classical dendroclimatology extracts climate signals by averaging out individual variation (top-down), while here **we model individual tree growth** (bottom-up)
- Climatic drivers capture inter-annual to decadal variability, but **systematically miss extreme growth shocks** that are likely due to individual endogenous processes [5]
- Accounting for these **non-linear, individual-level responses is critical for accurate forecasts** of forest vulnerability and carbon storage under climate change [6]

References

- [1] Gebrechorkos et al. Nature (2025)
- [2] Babst et al. Sci Adv (2019)
- [3] Gao et al. Nat Ecol Evol (2022)
- [4] Melvin & Briffa. Dendrochronologia (2008)
- [5] Grossiord et al. New Phytologist (2020)
- [6] Anderegg et al. Nat Clim Change (2020)